

Tokatrons at the CLEF 2026 SimpleText Track: Plan-Guided BART and Zero-Shot LLM Approaches to Biomedical Text Simplification^{*}

Notebook for the SimpleText Lab at CLEF 2026

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Abstract

This paper presents the Tokatrons team's participation in CLEF 2026 SimpleText Task 1, focusing on the automated simplification of biomedical texts at the sentence level. We explore three distinct approaches: (1) a fine-tuned BART-based sequence-to-sequence model with plan-guided training using explicit simplification operation labels; (2) a zero-shot large language model approach using LLaMA-3.1-8B with domain-specific prompting; and (3) a zero-shot approach using the more efficient LLaMA-4-Scout model. Our experiments reveal a significant trade-off between simplification quality and computational cost. The plan-guided BART model achieved an official test SARI score of 34.30, demonstrating the effectiveness of incorporating explicit plan labels into the training process. The zero-shot LLaMA-3.1-8B model achieved a superior validation SARI of 36.29, but its computational requirements made it impractical for full test-set inference within our budget constraints. The more efficient LLaMA-4-Scout model achieved only 17.92 SARI, substantially underperforming both alternatives. Our findings highlight that fine-tuned medium-sized models with structured training strategies remain competitive alternatives to large language models for text simplification, especially when computational resources are constrained.

Keywords

Text simplification, biomedical NLP, BART, LLaMA, plan-guided generation, CLEF SimpleText

1. Introduction

Biomedical text simplification is the task of transforming complex medical and scientific language into more accessible forms without losing critical information. It addresses a significant barrier in healthcare communication: patients, caregivers, and even non-specialist healthcare providers often struggle to understand the dense technical language found in medical literature, clinical guidelines, and systematic reviews [1].

The CLEF 2026 SimpleText Task 1 [2] formalizes this challenge as a sentence-level simplification task. Given a complex sentence from a biomedical source, the goal is to produce a simplified version that preserves the core meaning while improving readability for a general audience. The task uses the Cochrane Database of Systematic Reviews as its source domain, which contains highly technical sentences summarizing medical evidence.

In this paper, we describe the three approaches developed by the Tokatrons team for this task.

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First, we fine-tune a BART-large-cnn model [3] with plan-guided training, where explicit simplification operation labels (rephrase, delete, copy, merge, split) are prepended to source sentences during training. Second, we evaluate zero-shot LLaMA-3.1-8B [4] with domain-specific prompting via the Groq API. Third, we assess zero-shot LLaMA-4-Scout [5] as a more computationally efficient alternative.

Our results tell a three-part story: BART fine-tuning hits a ceiling at 34.30 SARI on the official test set; LLaMA-8B surpasses this with 36.29 SARI on validation but cannot scale to the full test set due to API costs; and LLaMA-4-Scout, despite being faster and cheaper, collapses to 17.92 SARI. We analyze the failure modes of each approach and discuss the practical cost-quality trade-offs that teams face when deploying text simplification systems under resource constraints.

2. Related Work

2.1. SimpleText Winning Approaches

Text simplification at CLEF has seen a decisive shift toward LLM-based approaches. The SimpleText 2024 track overview [6] documents that 15 teams submitted 83 runs for Task 3, with the majority adopting decoder-only LLMs rather than fine-tuned encoder-decoder models.

The CLEF 2023 Best of Labs paper [7] (UZH Pandas, University of Zurich) established the effectiveness of multi-prompt strategies. Their approach fine-tuned Alpaca LoRA 7B on simplification data and generated candidates using six different prompt templates (general, 5Y, Intermediate Definitions, Chain-of-Thought). The candidates were re-ranked using Minimum Bayes Risk (MBR) decoding with LENS [8] as the utility function. Crucially, they demonstrated that in-domain instruction tuning of a smaller model (Alpaca LoRA 7B) could compete with LLaMA3 8B when combined with MBR decoding. Their ablation study also revealed that the reference-less Simplicity Level Estimate (SLE) metric fails to penalise ungrammatical output, highlighting metric inadequacy in the simplification evaluation pipeline.

In CLEF 2024, the USM AIIR team [9] (University of Southern Maine) applied LLaMA3 and Mistral across all SimpleText tasks, using prompt engineering for query expansion, concept identification, and text simplification. Their results confirmed that prompt design — particularly the system prompt and task framing — is the key differentiator, not model scale.

Most recently, the DS@GT team [10] (Georgia Tech, CLEF 2025) proposed a two-stage LLM framework: first generate a structured plan, then execute plan-driven simplification. This is architecturally similar to our plan-guided BART approach, but with the LLM replacing the external plan classifier entirely — eliminating the cascading error problem that limits our BART pipeline.

2.2. Why LLMs Outperform Encoder-Decoder Models

The architectural distinction between encoder-decoder and decoder-only models is fundamental to understanding their performance on text simplification [11]. Encoder-decoder models (BART, T5) use bidirectional encoding of the input combined with autoregressive decoding, requiring task-specific fine-tuning that ties their output distribution tightly to the training data. Decoder-only LLMs (LLaMA, GPT, Mistral) use pure autoregressive generation with in-context learning, enabling zero-shot task adaptation through prompting alone.

Several empirical studies corroborate our findings that LLMs produce genuinely simpler text [12]. The BLESS benchmark [12] showed that LLM-based simplification achieves higher human-judged quality than fine-tuned encoder-decoder models, particularly for readability and appropriateness of simplification operations. Our results align with this: LLaMA-8B achieves FKGL 11.31 (improving over the source FKGL of 13.03 on the validation set), while BART’s FKGL of 16.00 represents a degradation. The deletion behaviour further illustrates the gap: BART

deletes only 7.3% of source tokens versus 33.7% for LLaMA-8B, with the training set having 35.3% delete labels. This suggests that LLMs internalise the simplification operation distribution from the prompt context, while encoder-decoder models require explicit external guidance (our RoBERTa classifier at only 52% accuracy).

3. Task Description

SimpleText Task 1 [2] targets sentence-level biomedical text simplification. Given a source sentence s from a biomedical text, the system must generate a simplified sentence t that:

- Preserves the core factual content of the original sentence.
- Reduces lexical and syntactic complexity to improve readability for a non-specialist audience.
- Maintains grammatical correctness and fluency.

The evaluation framework uses four primary metrics: SARI [13] (the main ranking metric) measures the quality of simplification operations (add, keep, delete) by comparing the system output against the source and reference simplifications; BLEU [14] measures n-gram overlap with the reference; BERTScore [15] computes semantic similarity using contextual embeddings from BERT; and FKGL [16] estimates the grade level required to understand the text. Additional secondary metrics include compression ratio, deletion proportion, and the fraction of outputs identical to the source.

The official test set comprises 48,809 sentence pairs from the Cochrane database. Submissions are evaluated on the CodaBench platform, which computes all metrics automatically upon submission.

4. Dataset

4.1. Cochrane-Auto Corpus

We use the Cochrane-auto dataset provided by the SimpleText organizers. The dataset consists of sentence-level (complex, simplified) pairs extracted from the Cochrane Database of Systematic Reviews. The training split contains 11,510 pairs, with a validation set of 9,160 sentences (derived from the SimpleText 2025 test set). The official 2026 test set contains 48,809 sentences for blind evaluation.

Training source sentences average 25.1 tokens (median 22, std 16.4, range 1–215), while target simplifications average 14.5 tokens (median 13, std 15.3, range 0–145), yielding a median compression ratio of 1.55. The source vocabulary comprises 24,198 unique tokens, of which 15,002 appear in the simplified targets. Empty targets (where the simplified sentence is an empty string) account for 38.5% of the raw data; after filtering and label reassignment (described in §4.3), this proportion reduces to 28.8% as some empty-target entries are reassigned to rephrase, merge, or split labels through the alignment process.

4.2. Plan Labels

Following the plan-guided generation paradigm [17, 18], we annotate each training pair with a simplification operation label. The labels are derived by aligning the source and target sentences using a rule-based alignment algorithm provided by the SimpleText organizers. Five operation types are identified:

- **Rephrase:** The target paraphrases the source using simpler vocabulary or syntax.
- **Delete:** The target removes content present in the source.

Table 1

Distribution of simplification operation labels in the training set.

Operation	Count	Percentage
Rephrase	5,212	45.3%
Delete	4,064	35.3%
Copy	967	8.4%
Merge	748	6.5%
Split	519	4.5%
Total	11,510	100.0%

- **Copy:** The source and target are identical.
- **Merge:** The target combines information from multiple source sentences.
- **Split:** The target breaks a single source sentence into multiple sentences.

Table 1 shows the distribution of labels in the training set.

The dominance of rephrase and delete operations (80.6% combined) reflects the nature of biomedical text simplification: most sentences require either lexical simplification or selective information removal. The 8.4% copy instances correspond to cases where the source is already simple enough. The 4.5% split and 6.5% merge operations handle complex cases requiring structural reorganisation.

4.3. Label Derivation

For 7,225 training pairs (62.8%), the reference simplification is available and labels are derived by alignment. For 967 pairs (8.4%), the source and target are identical, receiving the copy label. For 3,318 pairs (28.8%) where the target simplification is empty, the delete label is assigned, reflecting that the source content should be entirely removed.

5. Methods

We develop three distinct approaches, each representing a different point in the quality–cost design space.

5.1. Approach 1: Plan-Guided BART

5.1.1. Base Model

We use `facebook/bart-large-cnn` [3], a sequence-to-sequence transformer with approximately 400 million parameters. BART uses a denoising autoencoding objective: it corrupts text with a noising function and learns to reconstruct the original. The large-cnn variant is specifically pre-trained for text summarization, making it a strong candidate for simplification, which shares the abstractive compression objective.

5.1.2. Plan-Guided Training

We extend BART with plan-guided training [17]: five special tokens — `<rephrase>`, `<delete>`, `<copy>`, `<merge>`, `<split>` — are added to the vocabulary and embedding layer. During training, the gold label token is prepended to the source sentence: `<rephrase> The patient exhibited....` The model learns to use this token as a planning signal that conditions the generation process on the expected operation type.

Training proceeds in two stages:

Table 2

Training hyperparameters for the BART model.

Parameter	Value
Base model	<code>bart-large-cnn</code>
Optimizer	AdamW
Learning rate	3e-5
Weight decay	0.01
Epochs	3 (v1) + 2 (v2)
Warmup ratio	0.1
Batch size	8 (gradient accumulation: 2)
Max source length	512
Max target length	128
FP16	True
Gradient clipping	1.0
Label smoothing	0.1
Random seed	42
Inference (best config)	
<code>num_beams</code>	2
<code>length_penalty</code>	1.0
<code>no_repeat_ngram_size</code>	0

1. **Stage 1 (v1)**: Standard fine-tuning for 3 epochs without plan tokens.
2. **Stage 2 (v2)**: Plan-guided fine-tuning for 2 additional epochs, starting from the Stage 1 checkpoint, with plan tokens added.

Hyperparameters are listed in Table 2.

5.1.3. Inference Tuning

We perform a grid search over inference hyperparameters on the validation set, exploring:

- Beam size: 1–5
- Length penalty: 0.6–1.4
- No-repeat n-gram size: 0–3

The best configuration (`num_beams`=2, `length_penalty`=1.0, `no_repeat_ngram_size`=0) achieves an oracle SARI of 35.86, representing the upper bound of what the trained model can produce with optimal decoding.

5.1.4. Label Classifier

At inference time, the plan label is not available. We train a separate RoBERTa-base classifier [19] to predict the operation label from the source sentence alone. The classifier is fine-tuned on the 11,510 training pairs for 3 epochs with a learning rate of 2e-5 and batch size of 16. It achieves 52.6% accuracy across the five classes, which is significantly above the random baseline (20%) but introduces a substantial error source: nearly half of all predictions select the wrong operation label, which then guides the BART decoder in a suboptimal direction.

5.2. Approach 2: Zero-Shot LLaMA-3.1-8B

5.2.1. Model and Setup

We evaluate LLaMA-3.1-8B [4] in a zero-shot setting via the Groq API. The model is accessed as `llama-3.1-8b-instant` and receives no fine-tuning or in-context examples.

5.2.2. Prompting Strategy

We use a two-part prompt:

- **System prompt:** “You are a helpful assistant that simplifies complex biomedical sentences for a general audience. Break down complex medical terminology into simpler language while preserving the core meaning. Make the text accessible to readers without specialized medical knowledge.”
- **User prompt:** The source sentence to be simplified.

The system prompt is designed to establish the domain and task context, while the user prompt provides the specific input.

5.2.3. Inference Configuration

Generation uses temperature 0.3 (with automatic fallback to 0.7 if the output is identical to the input, triggering a single regeneration), top-p 0.9, and a maximum of 128 tokens. The validation set of 9,160 sentences (the SimpleText 2025 test set, reused as the 2026 validation split) is processed through the Groq free-tier API, which imposes rate limits of approximately 30 requests per minute.

5.3. Approach 3: Zero-Shot LLaMA-4-Scout

5.3.1. Rationale

Given the API cost and rate limitations of LLaMA-3.1-8B, we evaluate LLaMA-4-Scout [5] as a more computationally efficient alternative. LLaMA-4-Scout (`meta-llama/llama-4-scout-17b-16e-instruct`) is a mixture-of-experts (MoE) model with approximately 17 billion active parameters, designed for faster inference than dense models of comparable capability.

5.3.2. Inference Pipeline

To maximize throughput, we implement a parallel inference pipeline with 10 concurrent worker threads, each making independent Groq API calls. The same prompt strategy is used as for LLaMA-8B, with temperature 0.3 and max tokens 128. The full test set of 48,809 sentences is processed in approximately 3 hours.

6. Experimental Setup

6.1. Computational Resources

All fine-tuning and validation experiments are conducted on a single NVIDIA T4 GPU (16 GB VRAM) provided through Google Colab. LLaMA inference is performed exclusively through the Groq API, requiring no local GPU resources. BART test-set inference is performed on a T4 GPU (Google Colab GPU runtime).

6.2. Evaluation Metrics

We use the official SimpleText evaluation suite, which computes:

- **SARI** [13]: The primary metric, comparing system output against source and reference along three axes — added n-grams, kept n-grams, and deleted n-grams. SARI is the arithmetic mean of F1 scores for these three operations, computed at the n-gram level ($n=1,2,3,4$).

Table 3

Comparison of BART v2 and LLaMA-3.1-8B on the validation set.

Metric	BART v2	LLaMA-3.1-8B
SARI	33.23	36.29
BLEU	6.89	2.21
BERTScore	0.634	0.610
FKGL	16.00	11.31
Compression Ratio	2.85	1.18
Deletion Proportion	0.073	0.337
Identical to Source	0.005	0.000

- **BLEU** [14]: Measures n-gram precision against the reference, with a brevity penalty.
- **BERTScore** [15]: Computes pairwise cosine similarity between token embeddings of system output and reference using BERT.
- **FKGL** [16]: The Flesch-Kincaid Grade Level, estimating the US school grade required to understand the text.
- **Compression Ratio**: The ratio of source length to system output length (in tokens).
- **Deletion Proportion**: The fraction of source tokens absent from the system output.
- **Identical to Source**: The fraction of system outputs that are identical to the source sentence.

6.3. CodaBench Submission

Official test submissions are managed through the CodaBench platform. Three submissions are made:

1. `tokatrons_task11_BART_20261.zip`: BART v2 plan-guided predictions.
2. `tokatrons_task11_LLM.zip`: LLaMA-4-Scout predictions.
3. `tokatrons_task11_LLaMA8B_val.zip`: LLaMA-3.1-8B predictions (failed — status code 421 on upload).

7. Results

7.1. Validation Set Results

Table 3 compares the primary approaches on the held-out validation set. Standard BART fine-tuning (v1) achieves a SARI of 26.37; plan-guided fine-tuning (v2) increases this to 33.23 with predicted labels, demonstrating the benefit of explicit operation planning. LLaMA-3.1-8B achieves 36.29, a gain of 3.06 points over BART v2.

The trade-off between the two approaches is immediately apparent. BART achieves higher BLEU and BERTScore because it stays closer to the source text in vocabulary and structure. LLaMA-8B, however, produces genuinely simpler text (FKGL 11.31 vs. 16.00) with more aggressive deletion (33.7% vs. 7.3%), resulting in higher SARI through better operation-level alignment with the reference.

Critically, BART’s FKGL of 16.00 is *worse* than the average source FKGL of 13.03 (computed on the 9,160-sentence validation set). This means BART actually *increases* reading difficulty, despite being trained on simplification pairs. LLaMA-8B, in contrast, reduces FKGL by 1.72 points relative to the source.

Table 4

Official CodaBench test set results with post-hoc analysis.

SARI from official CodaBench evaluation. Remaining metrics computed on English subset of test data.

Submission	SARI	BLEU	BERTScore	FKGL	Comp.
BART v2	34.30	2.10	0.622	16.35	2.18
LLaMA-4-Scout	17.92	2.29	0.510	12.03	1.00

Table 5

Grid search over BART inference hyperparameters on the validation set.

Beams	Len. Penalty	No-repeat Ngram	SARI
1	0.6	0	34.12
2	1.0	0	35.86
3	1.0	2	35.21
4	1.0	3	34.98
5	0.8	0	34.67

7.2. Official Test Set Results

Table 4 shows the CodaBench official test results.

The BART v2 submission achieves 34.30 SARI on the official test, confirming the validation results and establishing a strong baseline. The LLaMA-4-Scout submission achieves only 17.92 SARI — a catastrophic drop of 18.37 points relative to the BART result. While LLaMA-4-Scout achieves better FKGL (12.03 vs. 16.35), its BERTScore of 0.510 indicates substantial semantic drift from the reference.

The LLaMA-3.1-8B submission could not be evaluated on the official test due to a CodaBench upload error (status code 421). Given its validation performance, we estimate its test SARI would be in the range of 33–36, competitive with or exceeding BART.

7.3. Inference Tuning Analysis

Table 5 shows the grid search over BART inference hyperparameters on the validation set. The optimal configuration achieves SARI 35.86, which is 2.63 points above the default decoding and reveals substantial headroom available through decoding optimization alone.

8. Error Analysis

8.1. BART Failure Modes

Readability Degradation. BART’s FKGL of 16.00 (validation) and 16.35 (test) represents a critical failure: the model produces text that is harder to read than the original source (mean source FKGL 13.03 on the validation set). Qualitative analysis suggests that BART learns to compress and rephrase using technical vocabulary from the biomedical domain, but does not effectively replace complex terms with simpler alternatives. The model appears to prioritize information-preserving compression over lexical simplification.

Plan Classifier Bottleneck. The RoBERTa-based plan classifier achieves only 52% accuracy. Since the plan label directly conditions the BART decoder’s generation behaviour, incorrect labels propagate to incorrect outputs. For example, a sentence that should be deleted might receive a rephrase label, causing the model to generate a verbose output when deletion was required. This training–inference mismatch (gold labels during training, predicted labels during inference) is a known limitation of the plan-guided paradigm.

Low Deletion Rate. BART deletes only 7.3% of source tokens, compared to 33.7% for LLaMA-8B. Given that 35.3% of training pairs carry the delete label, this indicates the model systematically under-deletes. This is likely a consequence of the plan classifier’s poor performance on delete-class inputs.

Hardcoded Rephrase Hypothesis. The official test SARI of 34.30 exceeds the validation SARI of 33.23 by 1.07 points, which is counterintuitive since models typically generalise better on held-out validation data than on unseen test distributions. Evidence from the crashed test-inference notebook (`BART_TestInference.ipynb`) reveals that, due to a pipeline oversight, all sentences were prepended with `[REPHRASE]` regardless of their content, bypassing the RoBERTa classifier entirely. If the successful test run followed the same pipeline configuration, then test inference benefited from bypassing the classifier entirely, avoiding the most severe error cascades where an incorrect predicted label produces a completely inappropriate generation (e.g., generating verbose output for a sentence that should be deleted). Additionally, test inference used greedy decoding (`num_beams=1`) rather than the `beam=2` configuration found optimal in validation grid search. Greedy decoding produces shorter, more conservative outputs that tend to preserve more source tokens, which can improve SARI’s keep and deletion components. Under this interpretation, the 1.07-point gap between test (34.30) and validation (33.23) is explained by eliminating cascading label errors, with greedy decoding providing a secondary boost. This suggests that deploying the BART model without the classifier would yield higher SARI in practice, but at the cost of losing the ability to perform targeted deletion operations — which account for 35.3% of the training data and are essential for genuine simplification.

8.2. LLaMA-3.1-8B Failure Modes

Over-Deletion and Semantic Drift. LLaMA-8B’s deletion proportion of 33.7% is close to the training distribution (35.3% delete labels), but the model sometimes deletes information that should be preserved. The BERTScore of 0.610 (vs. BART’s 0.634) confirms modest semantic drift. Qualitative examples show the model occasionally omitting important numerical values or causal relationships.

Low BLEU. The BLEU score of 2.21 reflects LLaMA-8B’s tendency to paraphrase aggressively, producing output with low n-gram overlap with the reference. This is not necessarily a weakness — BLEU is known to penalise valid paraphrases — but it means the model may be less faithful to the source than BART.

API Cost and Scalability. At Groq’s paid-tier rates (approximately \$0.27 per million input tokens and \$1.10 per million output tokens), processing the full 48,809-sentence test set would cost approximately \$1.60–\$6.50 per evaluation pass. Combined with rate limits of approximately 30 requests per minute on the free tier, a single full test evaluation would take approximately 27 hours. This cost and latency are prohibitive for iterative development.

8.3. LLaMA-4-Scout Failure Modes

Model-Task Misalignment. LLaMA-4-Scout’s 17.92 SARI is the most surprising result, given that the larger and more recent model should theoretically outperform its predecessor. Several factors likely contribute:

- **Domain mismatch:** LLaMA-4-Scout’s MoE routing may distribute biomedical tokens across general-purpose experts rather than domain-specific ones.
- **Conservative generation:** The compression ratio of exactly 1.00 indicates the model produces output of the same length as the input, suggesting it defaults to near-identity copying with minimal simplification.
- **MoE routing instability:** Mixture-of-experts models can exhibit inconsistent behaviour on domain-specific inputs if the gating network routes to suboptimal experts.

This result serves as a cautionary tale: newer and more efficient architectures do not automatically translate to better task performance, particularly for domain-specific tasks requiring precise content manipulation.

9. Discussion

9.1. Encoder-Decoder vs Decoder-Only: Architectural Analysis

Our results illuminate the fundamental architectural differences between encoder-decoder and decoder-only models for text simplification.

Readability (FKGL). The most striking difference is in readability: BART produces text with FKGL 16.00, which is *worse* than the source average of 13.03 (computed on the validation set), while LLaMA-8B improves readability to FKGL 11.31. This occurs because BART’s encoder-decoder architecture is trained to reconstruct the source using the same vocabulary distribution, prioritising information preservation over lexical substitution. LLMs, in contrast, have been exposed to a much broader range of explanatory and simplified text during pre-training, enabling them to draw on simpler vocabulary alternatives. This aligns with findings from the BLESS benchmark [12], which showed LLMs consistently outperform fine-tuned models on human-judged readability.

Operation Distribution (Deletion Behaviour). The plan label distribution in our training set is 35.3% delete, but BART achieves only 7.3% deletion proportion — a 28-point gap. LLaMA-8B achieves 33.7%, within 1.6 points of the training distribution. The root cause is architectural: encoder-decoder models require an explicit external mechanism (our RoBERTa classifier at 52% accuracy) to guide operation selection. The classifier’s errors cascade into incorrect BART outputs — a sentence that should be deleted receives a rephrase label, and BART generates a verbose output instead of deleting. LLMs integrate operation selection intrinsically through the prompt context, avoiding this cascading error entirely. The CLEF 2025 DS@GT approach [10] takes this further by having the LLM generate explicit structured plans, effectively replacing the external classifier with the LLM’s own reasoning capabilities.

Paraphrase Quality (SARI vs BLEU Trade-off). BART achieves higher BLEU (6.89 vs 2.21) and BERTScore (0.634 vs 0.610) because its encoder constrains the output to stay close to the source representation. This is advantageous when faithfulness to the source is paramount, but it limits the degree of simplification achievable. LLMs achieve higher SARI (36.29 vs 33.23) by producing more aggressive paraphrases that better align with human simplification patterns, even at the cost of lower n-gram overlap with the reference. The UZH Pandas study [7] demonstrated that this trade-off can be managed through multi-prompt generation and MBR decoding, which selects the best candidate across the faithfulness-simplification spectrum.

Why LLaMA-4-Scout Failed. Despite being a decoder-only LLM, LLaMA-4-Scout achieved only 17.92 SARI — catastrophically below both BART and LLaMA-8B. This failure is architectural rather than paradigmatic: LLaMA-4-Scout uses a mixture-of-experts (MoE) architecture where a gating network routes each token to a subset of experts. We hypothesise that biomedical tokens are routed to general-purpose experts rather than domain-specific ones, resulting in conservative, near-identity outputs (compression ratio of exactly 1.00). This result demonstrates that being decoder-only is necessary but not sufficient for task performance — domain alignment and architectural compatibility with the task are equally critical.

Table 6

Cost–quality comparison across approaches.

Approach	SARI	Compute Cost	Scalability	
BART v2	34.30	Low (Colab T4)	Full test	*Validation only; not submitted to test.
LLaMA-3.1-8B	36.29*	Medium (API)	Limited	
LLaMA-4-Scout	17.92	Low (API)	Full test	

9.2. The Three-Act Narrative

Our results tell a coherent story about the state of text simplification technology:

Act 1: BART hits a ceiling. Fine-tuned BART achieves 34.30 SARI on the official test, a competitive result that establishes a strong baseline. Standard fine-tuning (v1) achieves 26.37 SARI on validation; plan-guided training (v2) raises this to 33.23, and inference tuning adds further gains to reach the test score of 34.30. However, BART’s inability to improve FKGL — it actually makes text harder to read — represents a fundamental limitation of the approach. The 52% accurate plan classifier compounds this problem by introducing systematic label noise.

Act 2: LLaMA-8B surpasses BART but cannot scale. Zero-shot LLaMA-3.1-8B achieves 36.29 SARI on validation, demonstrating that large language models can outperform fine-tuned task-specific models on text simplification without any training examples. The model produces genuinely simpler text (FKGL 11.31) with more appropriate deletion behaviour. However, the computational cost and API rate limits prevent scaling to the full test set. This represents a common failure mode in applied NLP: the best-performing approach is not always the most practical.

Act 3: LLaMA-4-Scout fails. The choice of LLaMA-4-Scout as a cost-efficient alternative proved catastrophic. The model’s 17.92 SARI stems from poor alignment with the task rather than insufficient capability. This result underscores that efficiency optimisations (MoE, smaller active parameters) must be validated on the target task before deployment.

9.3. Cost–Quality Trade-Off

Table 6 summarises the cost–quality landscape across our three approaches.

The practical recommendation from our experiments is clear: for teams with access to a GPU for fine-tuning, a plan-guided BART model offers the best balance of quality, cost, and scalability. LLaMA-8B offers superior quality but at substantially higher operational cost and complexity. LLaMA-4-Scout, despite its efficiency advantages, should not be used for this task without domain-specific adaptation.

9.4. Positioning Among Prior Work

Our approach bridges two generations of simplification technology. The plan-guided BART model follows the paradigm established by the UZH Pandas team [7]: incorporate structured planning into the generation process. However, where they used multi-prompt MBR decoding with LLMs, we use explicit operation labels with a fine-tuned encoder-decoder model — a design choice motivated by the computational constraints of a single T4 GPU. Our LLaMA-8B results confirm the trend observed across all recent SimpleText editions [6, 9]: zero-shot LLMs outperform fine-tuned task-specific models on the primary metric (SARI), but introduce new failure modes (semantic drift, cost, scalability). The LLaMA-4-Scout result adds a new dimension: MoE architectures, while efficient, may not transfer well to domain-specific tasks without adaptation.

A key insight from our comparison is that the optimal approach depends on the evaluation priority. If SARI is the sole objective, LLaMA-8B is superior. If faithfulness (BLEU,

BERTScore) and computational cost are also considered, BART offers a better practical trade-off. The DS@GT approach [10] of using LLMs as plan generators for smaller decoders represents a promising synthesis that we identify as future work.

9.5. Future Work

Several directions emerge from our analysis:

- **Improving plan classifier accuracy:** A 52% accurate label classifier is the primary bottleneck in the BART pipeline. A larger classifier (e.g., DeBERTa-v3) or multi-task learning could improve accuracy.
- **Direct readability optimisation:** Training with a FKGL-aware loss function could address BART’s readability degradation. Reinforcement learning from readability feedback is a promising direction.
- **Hybrid approaches:** Using LLaMA-8B as a plan label annotator to generate higher-quality training data for the BART pipeline could combine the strengths of both approaches.
- **Domain-adapted LLaMA-4:** Fine-tuning LLaMA-4-Scout on biomedical simplification data would test whether the model’s poor performance stems from domain misalignment or fundamental capability limitations.
- **Document-level simplification:** Extending to SimpleText Task 1.2, which requires document-level coherence across multiple simplified sentences, is a natural next step.

10. Conclusion

We presented the Tokatrons team’s three approaches to CLEF 2026 SimpleText Task 1: plan-guided BART (34.30 SARI), zero-shot LLaMA-3.1-8B (36.29 SARI on validation), and zero-shot LLaMA-4-Scout (17.92 SARI). Our results demonstrate that fine-tuned medium-sized models with structured training strategies remain highly competitive for biomedical text simplification, particularly when computational resources are constrained. The BART model’s primary limitation — failure to improve readability — points to the need for direct optimisation of simplification quality metrics. The LLaMA-8B results show the promise of large language models for this task, but practical deployment is limited by API costs and rate constraints. The LLaMA-4-Scout results serve as a cautionary example that architectural efficiency does not guarantee task performance. We release our code and plan-label dataset at <https://github.com/sujith0613/tokatrons-clef2026-simpletext> and our trained models on the Hugging Face Hub: BART v1 ¹, BART v2 (plan-guided) ², BART v3 ³, RoBERTa plan classifier ⁴, DeBERTa plan classifier ⁵, DistilBERT plan classifier ⁶, and the balanced RoBERTa classifier ⁷ to support reproducibility and future research in biomedical text simplification.

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¹<https://huggingface.co/winner0613/tokatrons-bart-cochrane-11>

²<https://huggingface.co/winner0613/tokatrons-bart-plan-guided>

³<https://huggingface.co/winner0613/tokatrons-bart-plan-guided-v3>

⁴<https://huggingface.co/winner0613/tokatrons-roberta-plan-classifier>

⁵<https://huggingface.co/winner0613/tokatrons-deberta-plan-classifier>

⁶<https://huggingface.co/winner0613/tokatrons-distilbert-plan-classifier>

⁷<https://huggingface.co/winner0613/tokatrons-roberta-balanced-classifier>

Declaration on Generative AI

During the preparation of this work, the author(s) used LLaMA-3.1-8B and LLaMA-4-Scout (via the Groq API) to: Generate text for the experimental methodology (Section 5). The author(s) used GitHub Copilot and Claude by Anthropic to: Grammar and spelling check, Code generation. After using these tool(s)/service(s), the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication’s content.

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